

Milestone 6: AI-Powered Decision Support

Objective

The objective of this phase was to enhance the leave approval process by implementing AI-powered decision support mechanisms. The system analyzes leave requests and provides risk assessments and recommendations that assist managers in making informed approval decisions.

Overview

Traditional leave approval systems rely entirely on manual review by managers. In this project, AI-powered decision support was introduced to improve efficiency and consistency during the approval process.

The AI functionality evaluates leave requests based on predefined business rules and automatically generates:

- > Risk Assessment
- > Approval Recommendations

These insights help managers prioritize and review leave requests more effectively.

AI Risk Assessment

Description

The AI Risk Assessment feature evaluates leave requests based on the duration of leave and classifies them into different risk categories.

Risk Categories

Leave Duration	Risk Level
1 – 3 Days	Low Risk
4 – 7 Days	Medium Risk
More than 7 Days	High Risk

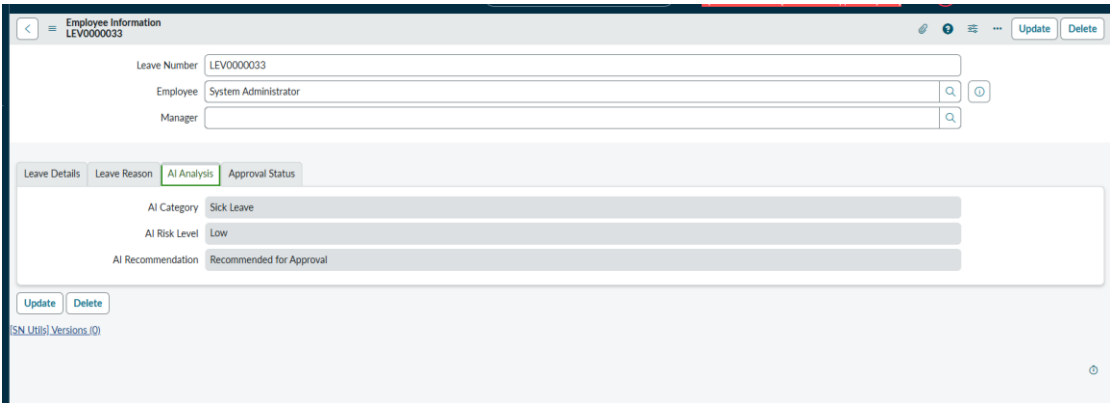
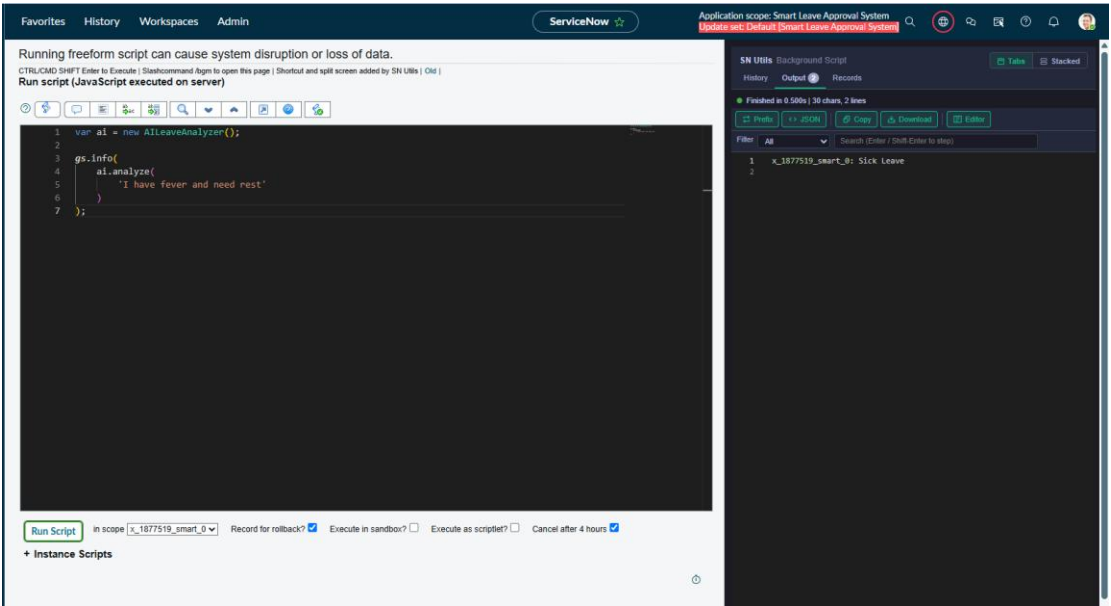
Low Risk

Criteria

- > Short-duration leave requests.
- > Minimal impact on business operations.

AI Output

Low Risk



Recommendation

- Leave request appears normal.

Medium Risk

Criteria

- >Moderate-duration leave requests.
- > Potential impact on team workload.

AI Output

Medium Risk

Employee Information
LEV0000032

Leave Number

LEV0000032

Employee

System Administrator

Manager

Leave Details

Leave Reason

AI Analysis

Approval Status

AI Category

AI Risk Level

Medium

AI Recommendation

Update

Delete

SN Utils | Versions (0)

Recommendation

Manager review recommended.

High Risk

Criteria

- > Extended leave duration.
- > Higher impact on operational activities.

AI Output

High Risk

Employee Information
LEV0000126

Leave Number

LEV0000126

Employee

System Administrator

Manager

Leave Details

Leave Reason

AI Analysis

Approval Status

AI Category

AI Risk Level

High

AI Recommendation

Manager Review Required

Update

Delete

Versions (0)

Recommendation

Extended leave requires additional review.

AI Recommendation Engine

Description

Based on the calculated risk level, the system automatically generates recommendations for managers.

Purpose

- 1. Improve approval consistency.
 - 2. Assist managers in decision-making.
 - 3. Reduce manual evaluation effort.
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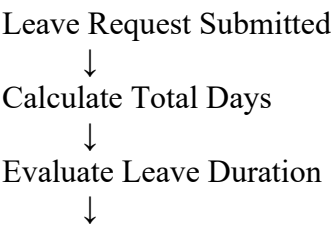
Sample Recommendations

Risk Level	Recommendation
Low	Leave request appears normal.
Medium	Manager review recommended.
High	Extended leave requires additional review.

AI Workflow Integration

The AI engine is integrated directly into the leave request lifecycle.

Process Flow



Generate AI Risk Assessment
↓
Generate AI Recommendation
↓
Store Results in Record
↓
Manager Review
